TTK4240 Industriell elektroteknikk

January 14, 2025

Marta Molinas
Institutt for teknisk kybernetikk

Lecture - Week 35 – Part II

Review of Low-Pass and High-Pass Filters

&

Instantaneous & Average Power in brief

- **Bandpass Filters**
  - Center Frequency, Bandwidth and Quality Factor
  - Series RLC
- **Bandreject Filters**
  - Series RLC
  - Transfer Function

Nilsson's Book Chapter 14: 14.1 to 14.5

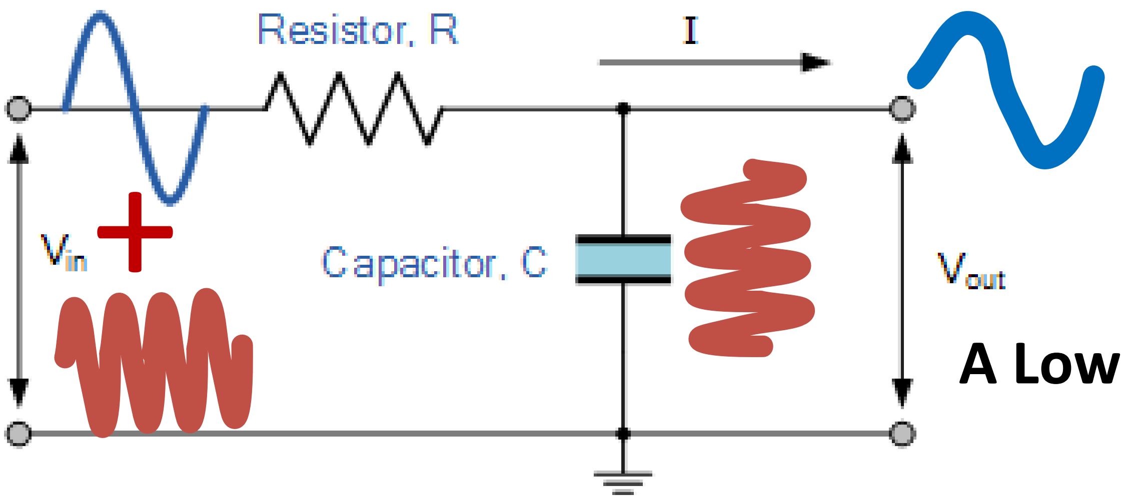
Review of the low-and high-pass Filters

Passive Filter

Review

Definition

The passive filter is a ***frequency-dependent circuit*** that can ***selectively pass*** signals at certain frequencies, while attenuates the signals at other frequencies.



A Low pass filter

Key point for filter design

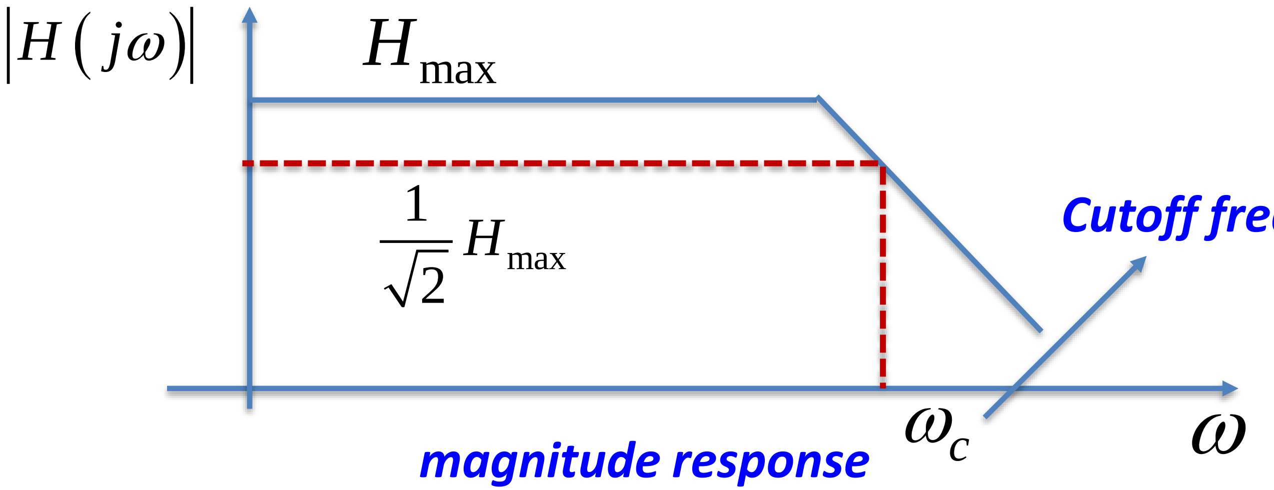
Review

We need some frequency-dependent ***circuit elements***, so that we can use their frequency-dependent ***properties*** to design a frequency ***selective circuit***.

L and C	$\omega = 0$ ***(Low frequency)***	$\omega = \infty$ ***(High frequency)***
$Z_L(j\omega) = j\omega L$	*L*  ***Pass***	*L*  ***Block***
$Z_C(j\omega) = \dfrac{1}{j\omega C}$	*C*  ***Block***	*C*  ***Pass***

The Cutoff Frequency ωc
for pratical passive filters

Review



- The cutoff frequency is the frequency at which the magnitude of the transfer function equals:

$$\left(1/\sqrt{2}\right)H_{max}$$

The physical meaning of cutoff frequency: *a power perspective* ?

Review

$$
\begin{aligned}
|V_L(j\omega_c)| &= |H(j\omega_c)||V_i| \\
&= \frac{1}{\sqrt{2}} H_{\max}|V_i| \\
&= \frac{1}{\sqrt{2}} V_{L\max}.
\end{aligned}
$$

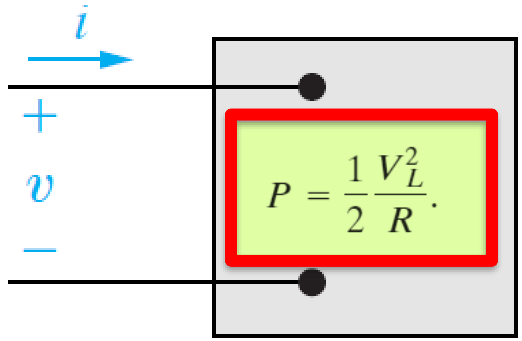
$$
\begin{aligned}
P(j\omega_c) &= \frac{1}{2}\frac{|V_L^2(j\omega_c)|}{R} \\
&= \frac{1}{2}\frac{\left(\frac{1}{\sqrt{2}}V_{L\max}\right)^2}{R} \\
&= \frac{1}{2}\frac{V_{L\max}^2/2}{R} \\
&= \boxed{\frac{P_{\max}}{2}.}
\end{aligned}
$$

- The Cutoff Frequency is the frequency at which the *power transfer to the output is reduced to half!*

Check 14.2 Defining the cutoff frequency, page 525

Instantaneous Power (10.1 Nilsso

Hand notes



$$v = V_m \cos(\omega t + \theta_v),$$

$$i = I_m \cos(\omega t + \theta_i),$$

Instantaneous power is defined as

$$p = vi.$$

$$p = V_m I_m \cos(\omega t + \theta_v) \cos(\omega t + \theta_i)$$

Check also 10.3 for a simpler calculation through RMS value, page 366

Use this Trigonometric Identity

$$\cos\alpha\cos\beta = \frac{1}{2}\cos(\alpha - \beta) + \frac{1}{2}\cos(\alpha + \beta)$$

$$p = V_m I_m \cos(\underbrace{\omega t + \theta_v}_{\alpha})\cos(\underbrace{\omega t + \theta_i}_{\beta})$$

$$p = \frac{V_m I_m}{2}\cos(\theta_v - \theta_i) + \frac{V_m I_m}{2}\cos(2\omega t + \theta_v + \theta_i)$$

Redefine the time at which the current phase crosses zero, i.e. $\omega t = \omega\tau - \theta_i$

$$p = \frac{V_m I_m}{2}\cos(\theta_v - \theta_i) + \frac{V_m I_m}{2}\cos(2\omega\tau + \theta_v - \theta_i)$$

$$\cos(\alpha + \beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$$

Resubstitute

$$\tau = t$$

$$p = \frac{V_m I_m}{2}\cos(\theta_v - \theta_i) + \frac{V_m I_m}{2}\cos(2\omega t + \theta_v - \theta_i)$$

$$p = \frac{V_m I_m}{2}\cos(\theta_v - \theta_i) + \frac{V_m I_m}{2}\cos(\theta_v - \theta_i)\cos(2\omega t) - \frac{V_m I_m}{2}\sin(\theta_v - \theta_i)\sin(2\omega t)$$

constant

sinusoidal

Average and Reactive Power

$$p = \frac{V_m I_m}{2}\cos(\theta_v - \theta_i) + \frac{V_m I_m}{2}\cos(\theta_v - \theta_i)\cos 2\omega t$$
$$- \frac{V_m I_m}{2}\sin(\theta_v - \theta_i)\sin 2\omega t.$$

Defined as P or average power

$$p = P + P\cos 2\omega t - Q\sin 2\omega t,$$

$$P = \frac{1}{T}\int_{t_0}^{t_0+T} p\,dt,$$ Average value

$$P = \frac{V_m I_m}{2}\cos(\theta_v - \theta_i),$$

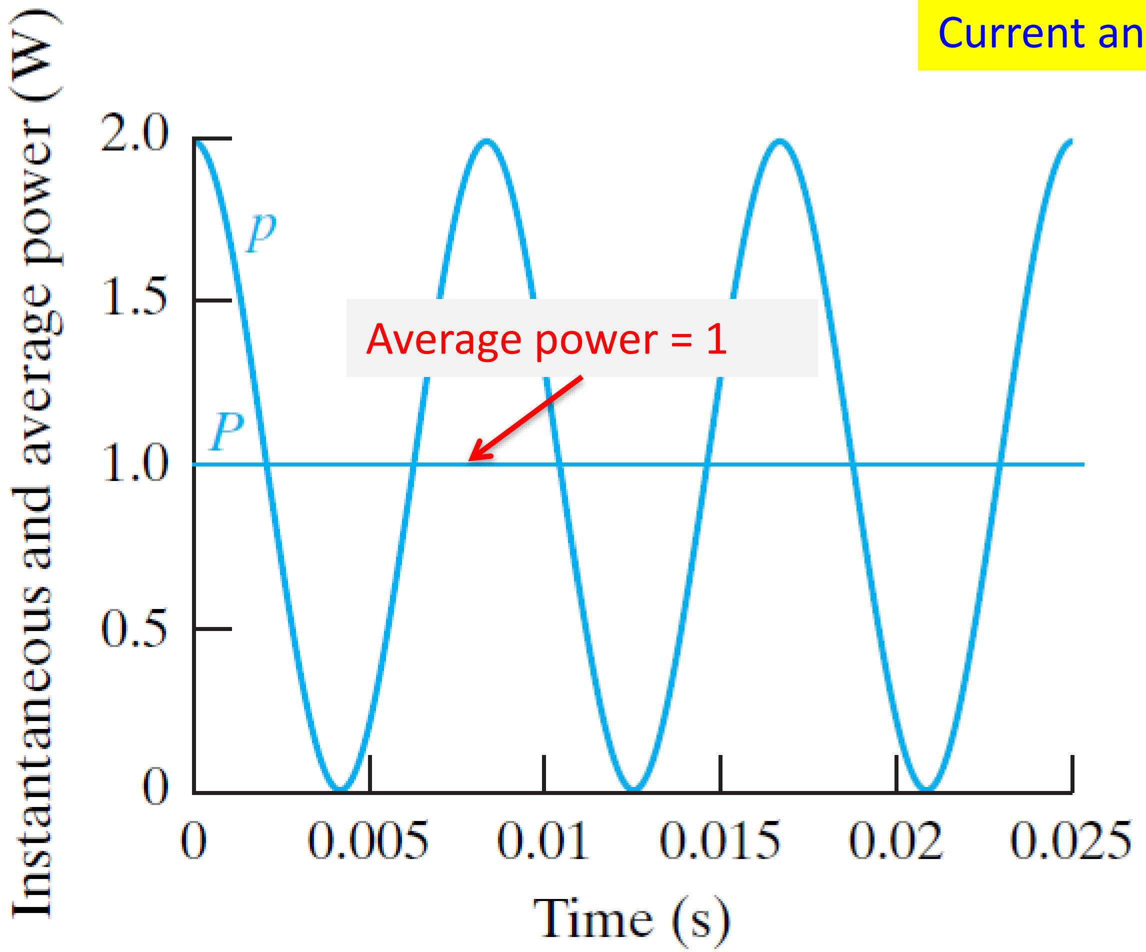
Active power

$$Q = \frac{V_m I_m}{2}\sin(\theta_v - \theta_i).$$

Reactive power

Power for purely resistive circuits

Current and voltage are in phase

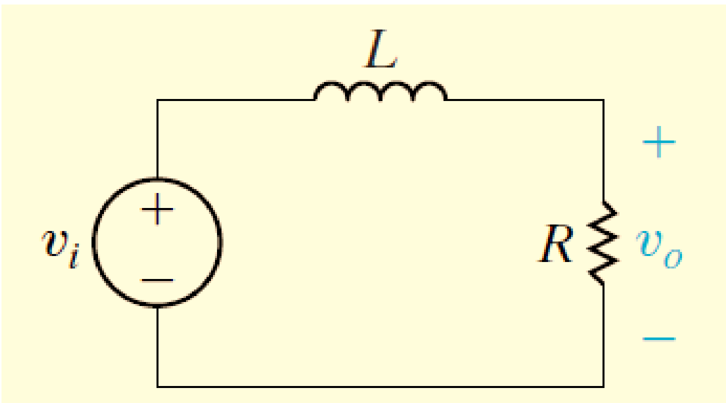


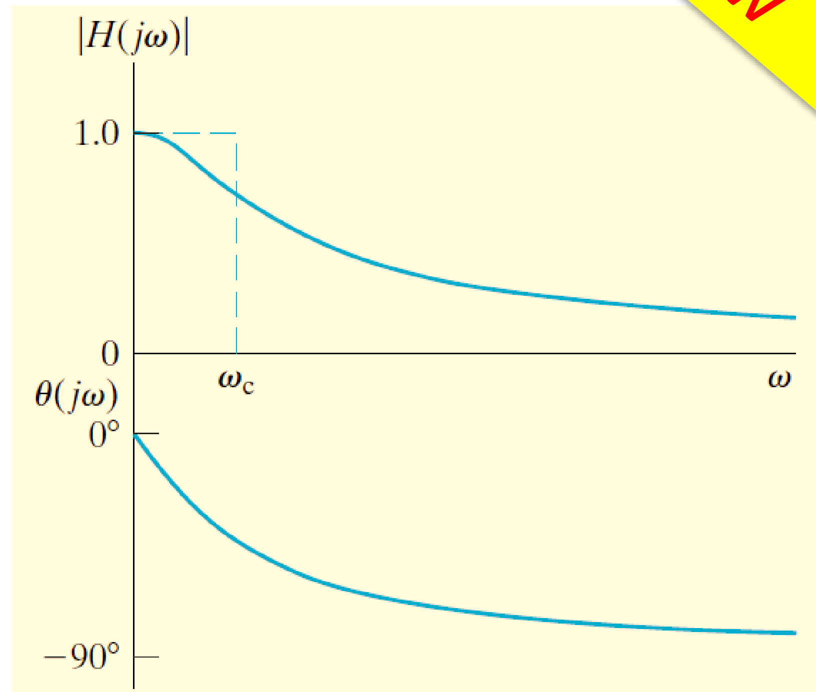
$$\theta_v = \theta_i.$$

$$p = P + P\cos 2\omega t.$$

Review

Low-Pass Filters: ***RL-based***





$$|H(j\omega)| = \frac{R/L}{\sqrt{\omega^2 + (R/L)^2}},$$

$$\theta(j\omega) = -\tan^{-1}\left(\frac{\omega L}{R}\right).$$

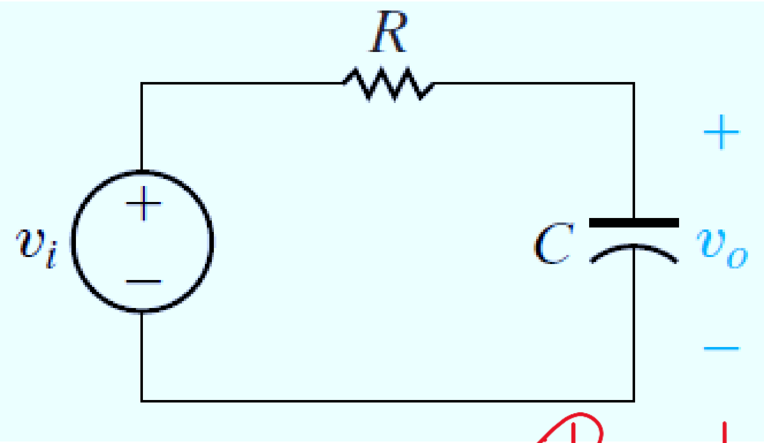
$= \frac{Im}{Re}$

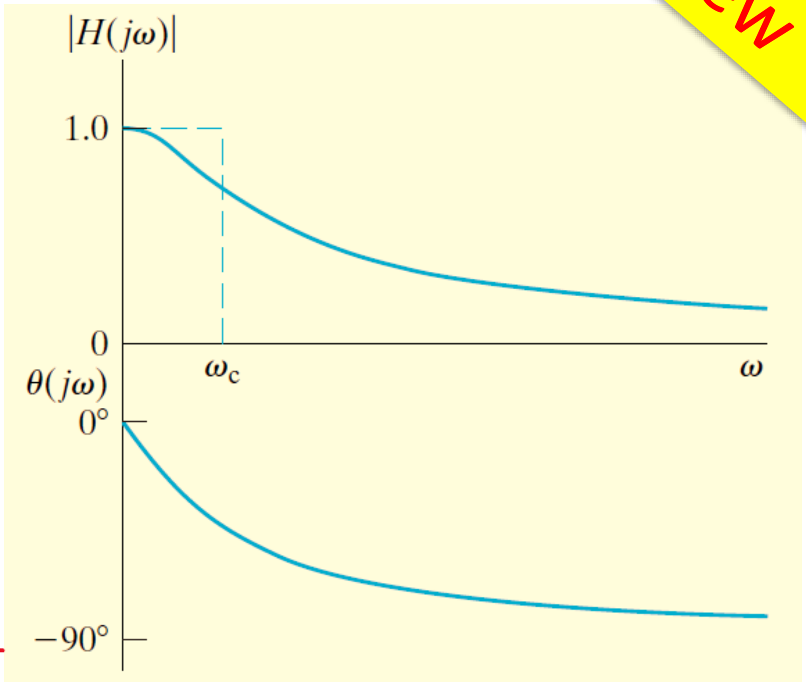
Cut off frequency

$$\omega_c = \frac{R}{L}$$

Low-Pass Filters: *RC-based*

Review





Real number

$$|H(j\omega)| = \frac{1/RC}{\sqrt{\omega^2 + (1/RC)^2}}$$

$$\theta(j\omega) = -\tan^{-1}\frac{\omega}{1/RC} = \frac{Im}{Re}$$

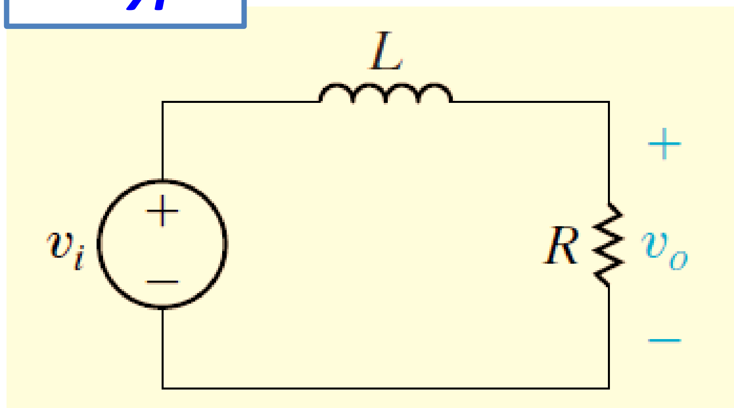
Cut off frequency

$$\omega_c = \frac{1}{RC}$$

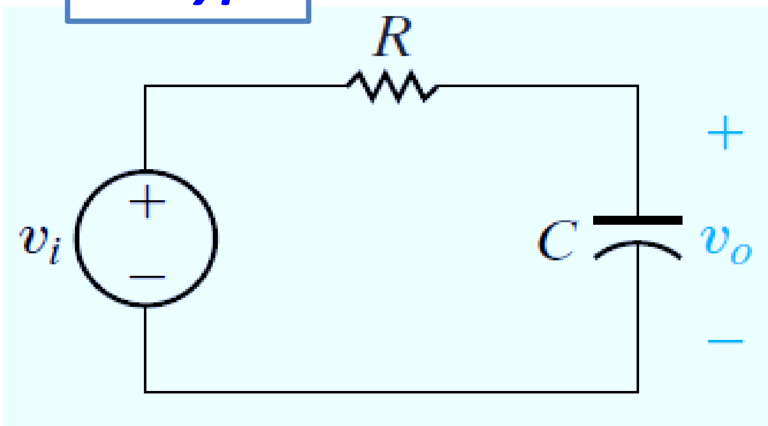
Low-pass filter comparison

Review

RL type



RC type



$$\omega_c = \frac{R}{L}$$

$$\omega_c = \frac{1}{RC}$$

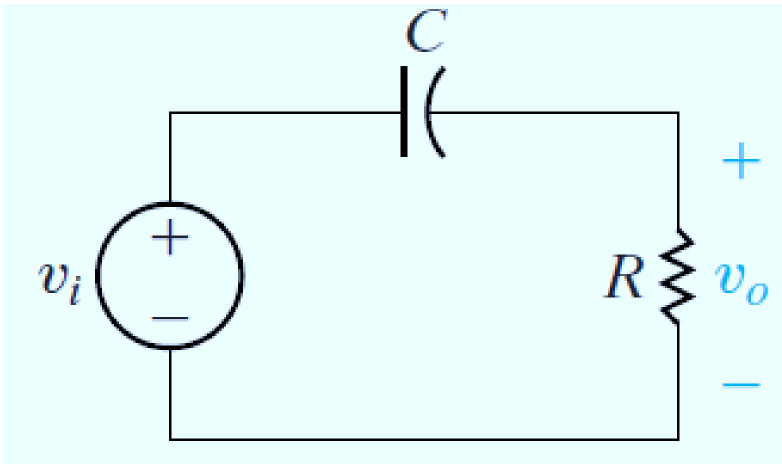
$$H(s) = \frac{\boxed{\omega_c}}{\boxed{s} + \omega_c}$$

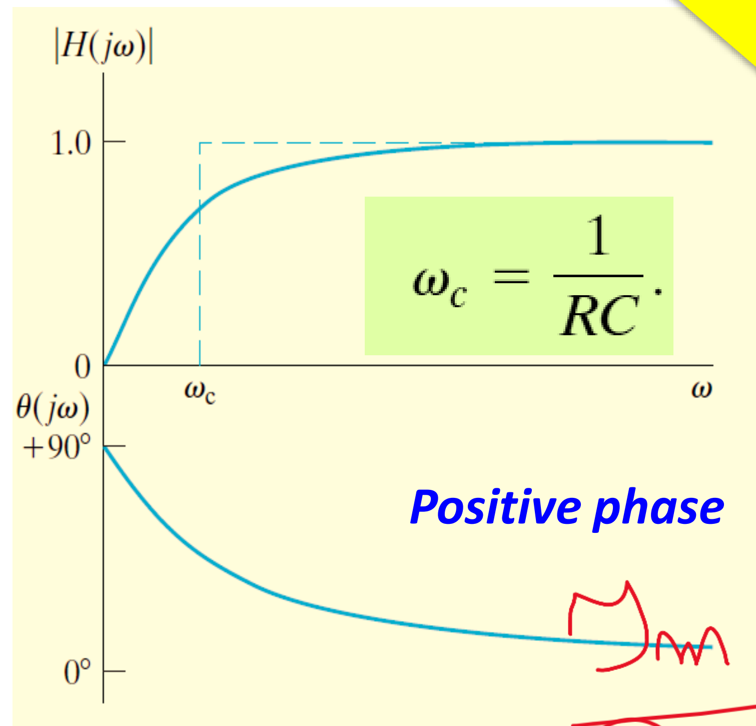
Maximum order of polynomials is ONE

First-order low pass filter

Review

High-Pass Filters: *RC type*



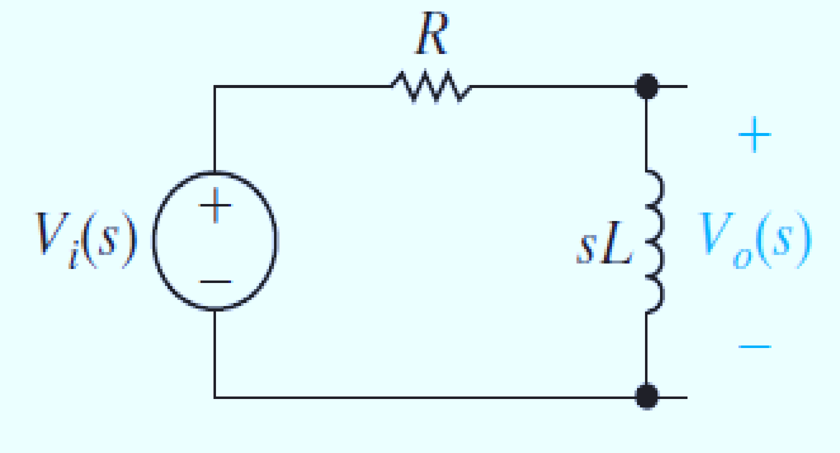


$$H(s) = \frac{V_o(s)}{V_i(s)} = \frac{s}{s + 1/RC}$$

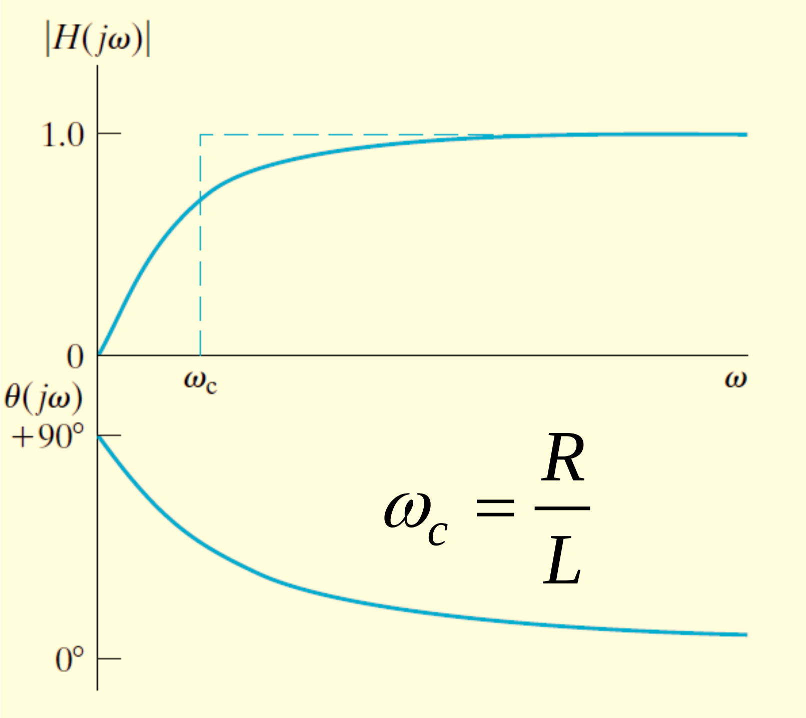
$$|H(j\omega)| = \frac{\omega}{\sqrt{\omega^2 + (1/RC)^2}},$$

$$\theta(j\omega) = 90^\circ - \tan^{-1}\omega RC.$$

High-Pass Filters: *RL type*



$$H(s) = \frac{V_o(s)}{V_i(s)} = \frac{sL}{R + sL}$$



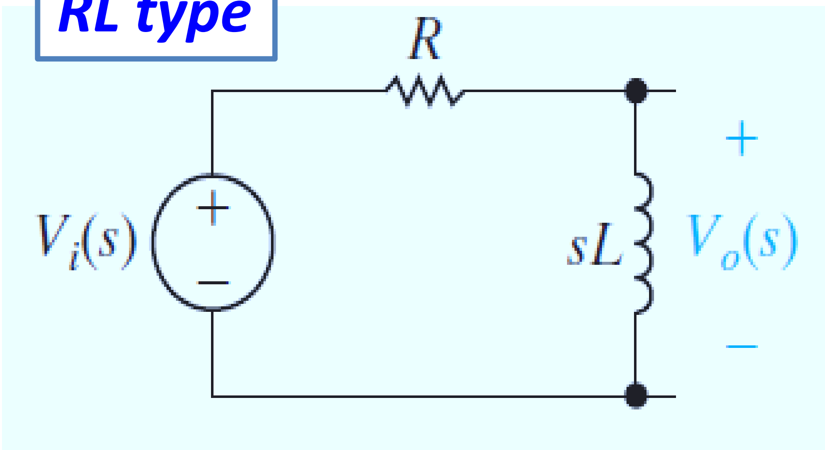
$$\omega_c = \frac{R}{L}$$

$$|H(j\omega)| = \frac{\omega}{\sqrt{(R/L)^2 + \omega^2}}$$

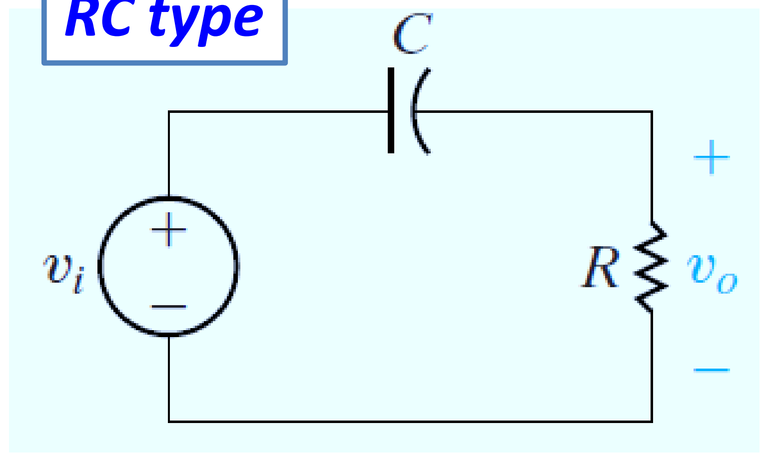
$$\theta(j\omega) = 90^\circ - \tan^{-1}\frac{\omega}{R/L}$$

High-pass filter comparison

RL type



RC type



$$\omega_c = \frac{R}{L}$$

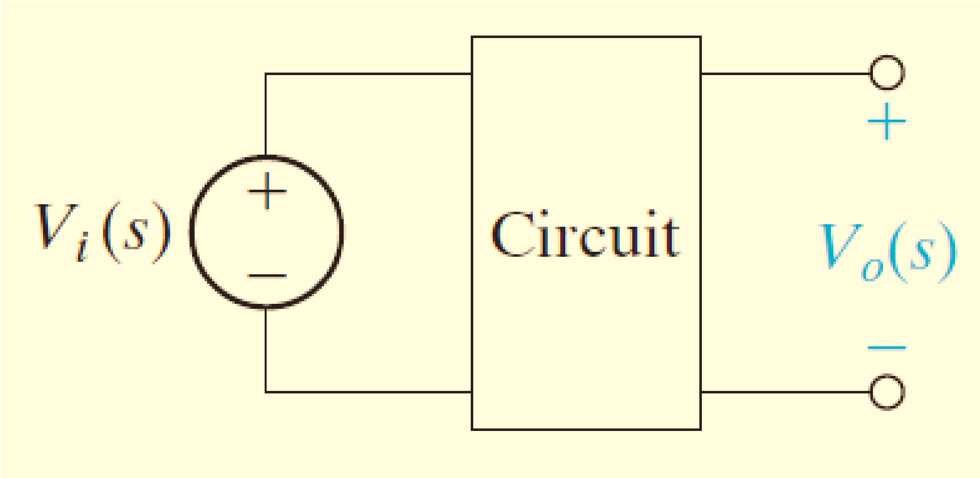
$$\omega_c = \frac{1}{RC}$$

$$H(s) = \frac{s}{\boxed{s} + \boxed{\omega_c}}$$

First-order high pass filter

Maximum order of polynomials is ONE

Loaded passive filters



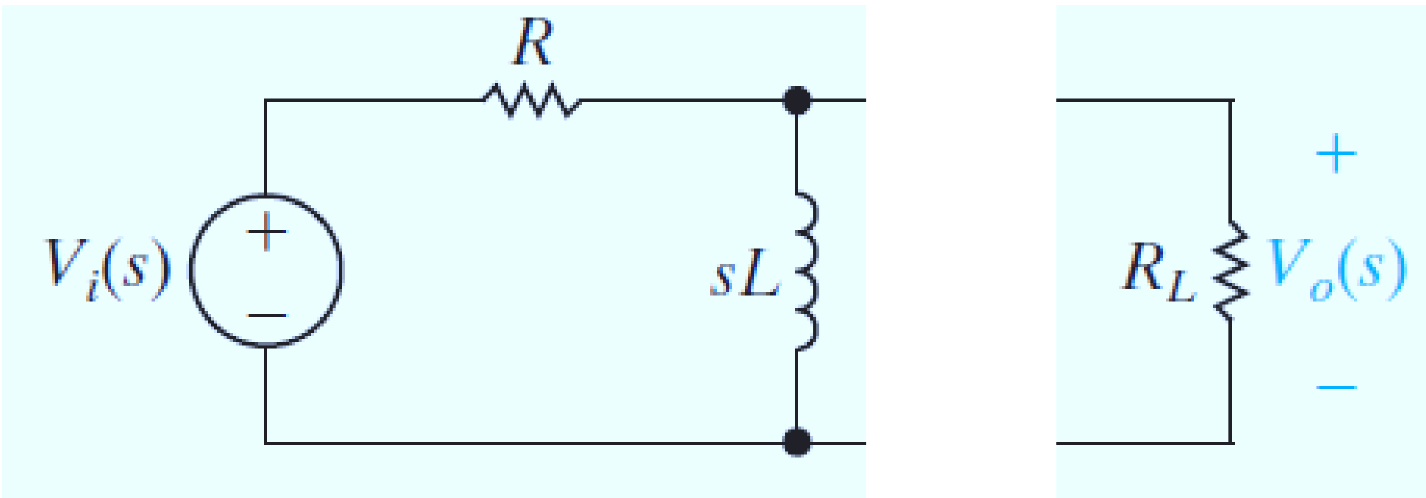
Problem:

If the filter's output is connected to other elements, like a resistive load, what will happen ?

And, will it affect the performance of the filter?

Example:

Ex. 14.3 & 14.4 (Nilsson)



$$V_{out}(s) = \frac{R_L // sL}{R + \boxed{R_L // sL}} V_{in}(s) \qquad R_L // sL = \frac{R_L \cdot sL}{R_L + sL}$$

Transfer function of the loaded filter

$$H(s) = \frac{V_{out}(s)}{V_{in}(s)} = \frac{s \boxed{\frac{R_L}{R_L + R}}}{s + \boxed{\frac{R}{L}\frac{R_L}{R_L + R}}}$$

$$K = \frac{R_L}{R_L + R}$$

$$\omega_c = K\frac{R}{L}$$

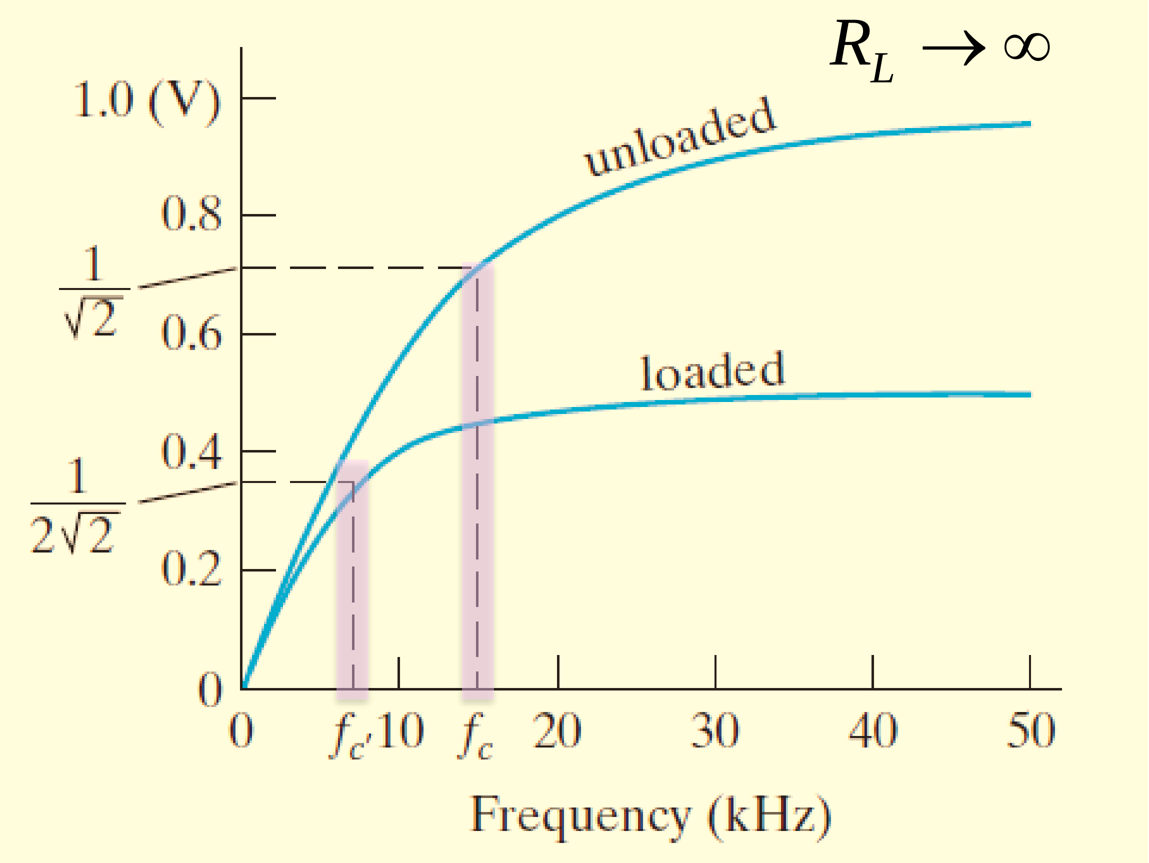
$$H(s) = \frac{sK}{s + \omega_c}$$

Frequency response

$$s = j\omega$$

$$\left|H(j\omega)\right| = \frac{\omega K}{\sqrt{\omega^2 + \omega_c^2}}$$

$$\angle H(j\omega) = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{\omega_c}$$

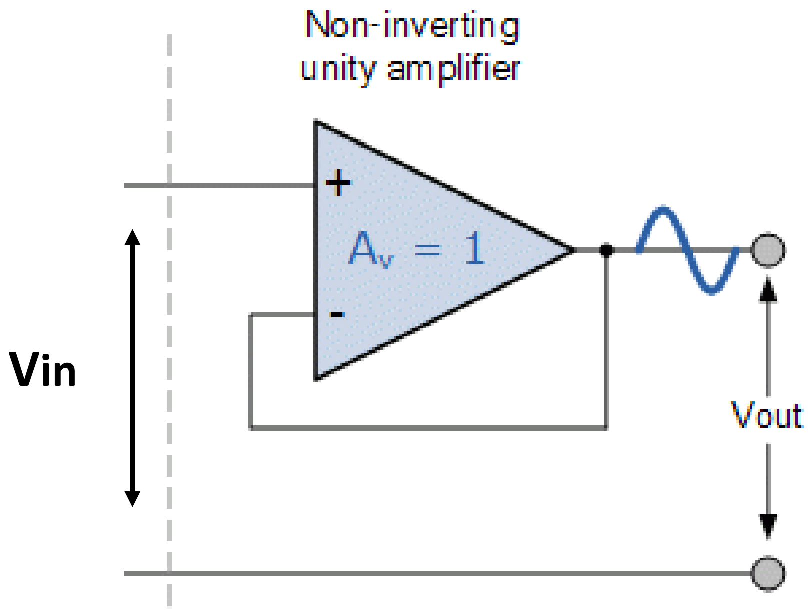


$$K = \frac{R_L}{R_L + R} \begin{cases} K < 1 \\ R_L \to \infty, K \approx 1 \end{cases}$$

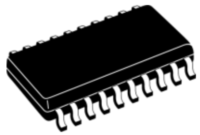
Output signals are degraded under the loaded filter

NTNU

Amplifier*



Amplifer

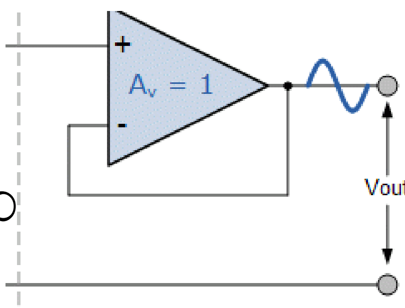


Two features

$$V_{in} \approx V_{out}$$

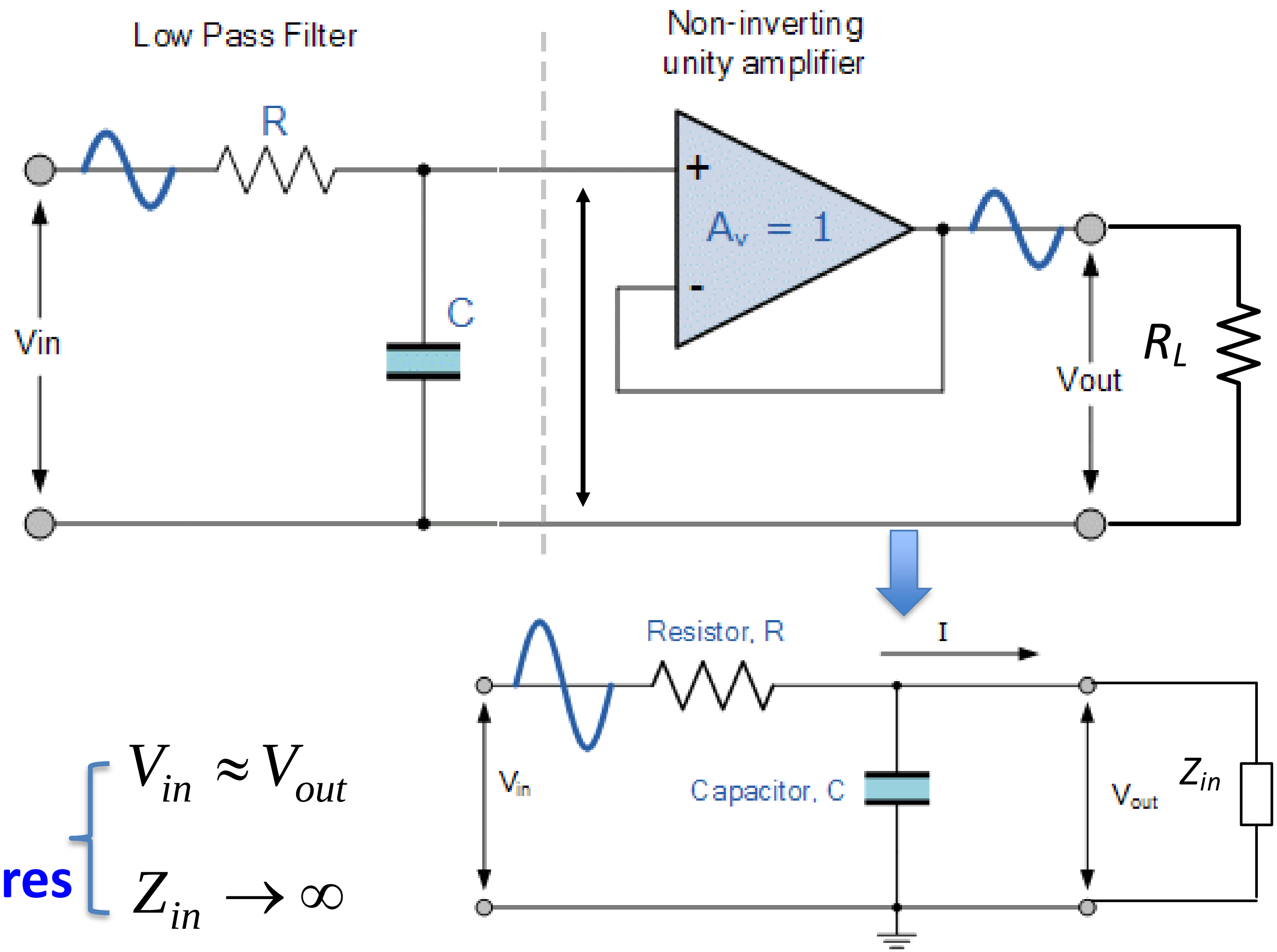
$$Z_{in} \to \infty$$

$$Z_{in} \to \infty$$



*supplementary contents

Active filters*



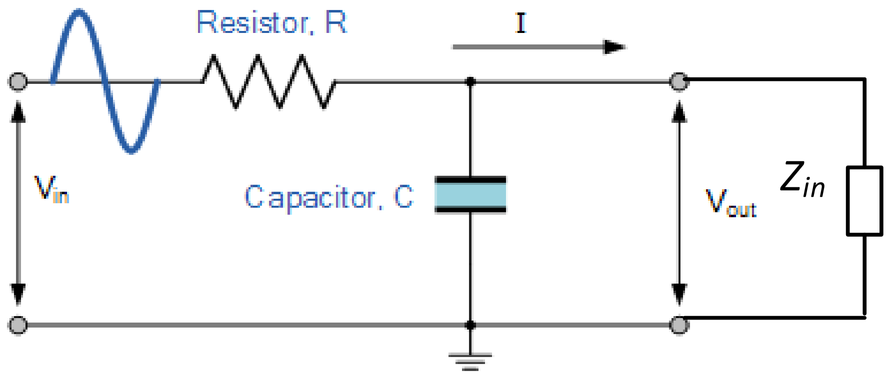
Two features

$$V_{in} \approx V_{out}$$

$$Z_{in} \rightarrow \infty$$

*supplementary contents

Active filter*



$$H(s) = \frac{V_{out}}{V_{in}} = \frac{Z_{in} // \frac{1}{sC}}{R + Z_{in} // \frac{1}{sC}} \approx \boxed{\frac{\frac{1}{sC}}{R + \frac{1}{sC}}}$$

The active filter can assure the performance of passive filter regardless of the **load variation**

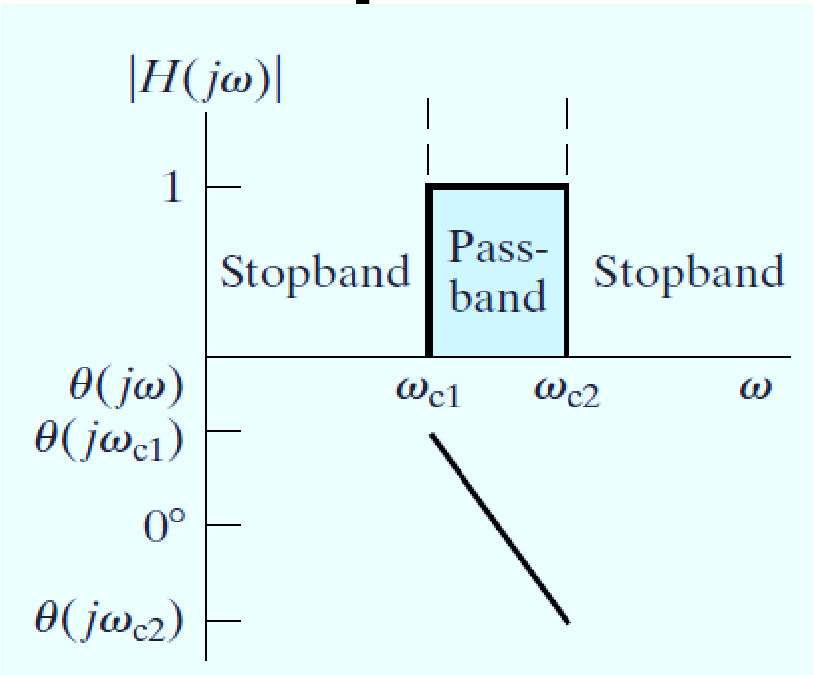
*supplementary contents

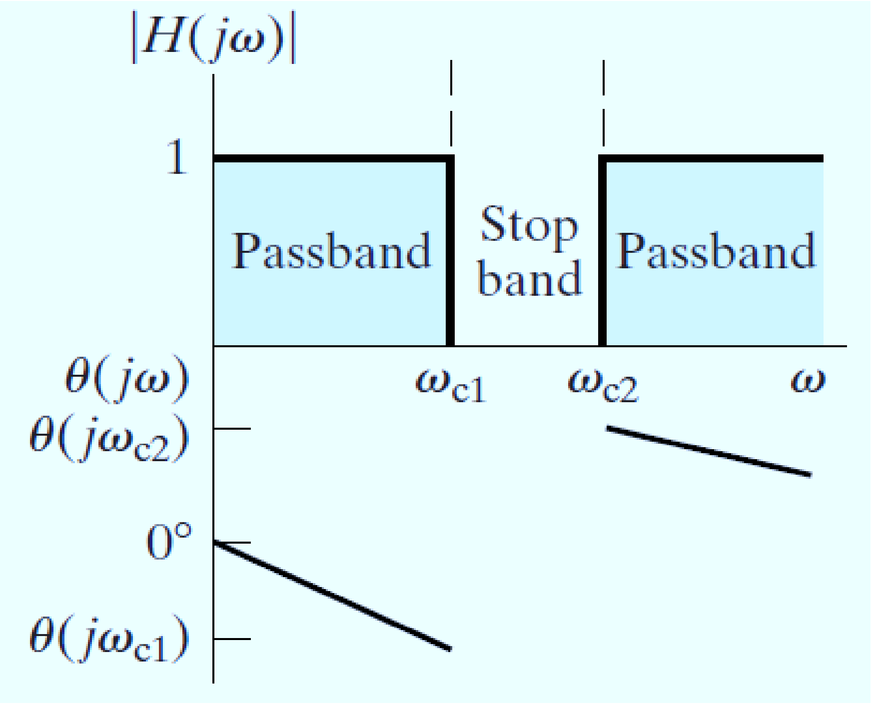
Break

Bandpass filter

Hand notes

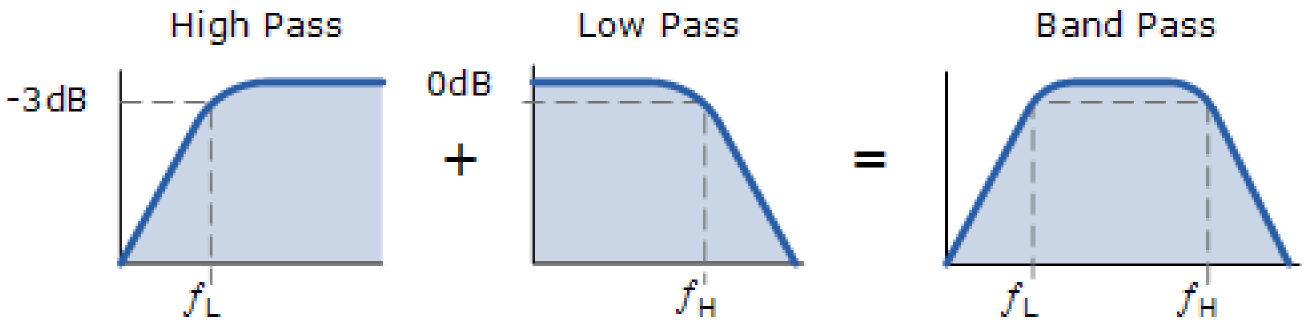
Band-pass or -reject filters





Shape of magnitude response changes TWICE

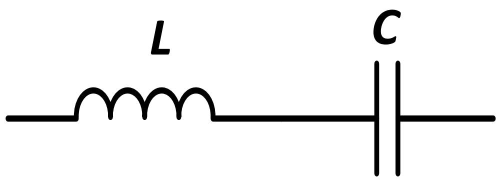
Implies we need TWO different types of frequency-dependent circuit elements to design such filters


High Pass
-3dB
f_L
+
Low Pass
0dB
f_H
=
Band Pass
f_L
f_H

Frequency-domain characteristics of ***series connected L and C***

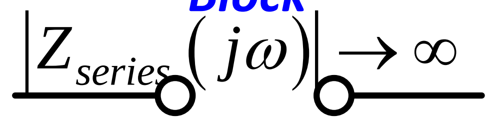
for both zero & infinity

Series connected



$\boxed{\omega = 0}$ $\boxed{\omega = \infty}$

Block



Frequency-domain impedance

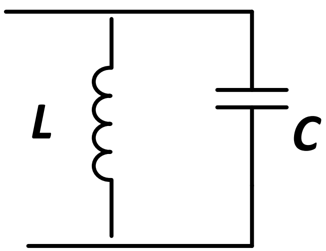
$$\underbrace{Z_{series}(j\omega) = j\omega L}_{Z_L} + \underbrace{\frac{1}{j\omega C}}_{Z_C}$$

$$Z_{series}(j\omega_0) = 0 \quad \Rightarrow \quad \boxed{\omega_0 = \sqrt{\frac{1}{LC}}}$$

Pass

Frequency-domain characteristics of *parallel connected L and C*

Parallel connected

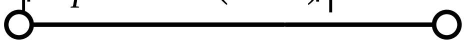


$\boxed{\omega = 0}$ $\boxed{\omega = \infty}$

$\left| Y_{parallel}(j\omega) \right| \to \infty$

$\left| Z_{parallel}(j\omega) \right| \to 0$

Pass



Frequency-domain admittance

$$Y_{parallel}\left(j\omega \right) = \underbrace{\frac{1}{j\omega L}}_{Y_L} + \underbrace{j\omega C}_{Y_C}$$

$Z_{parallel}\left(j\omega_0 \right) \to \infty$

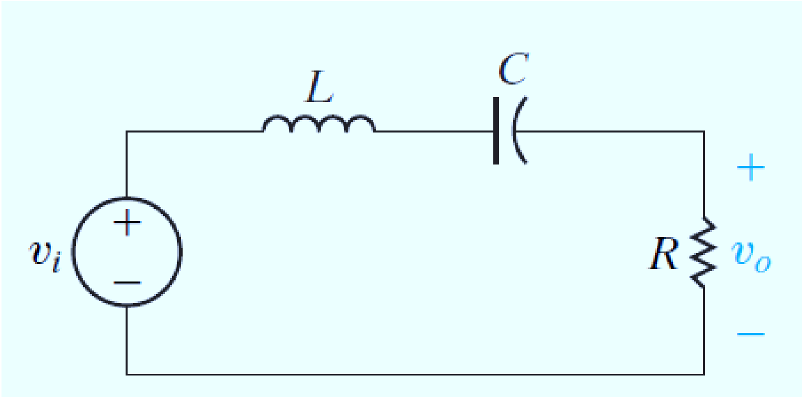
$Y_{parallel}\left(j\omega_0 \right) \to 0$

$$\boxed{\omega_0 = \sqrt{\frac{1}{LC}}}$$

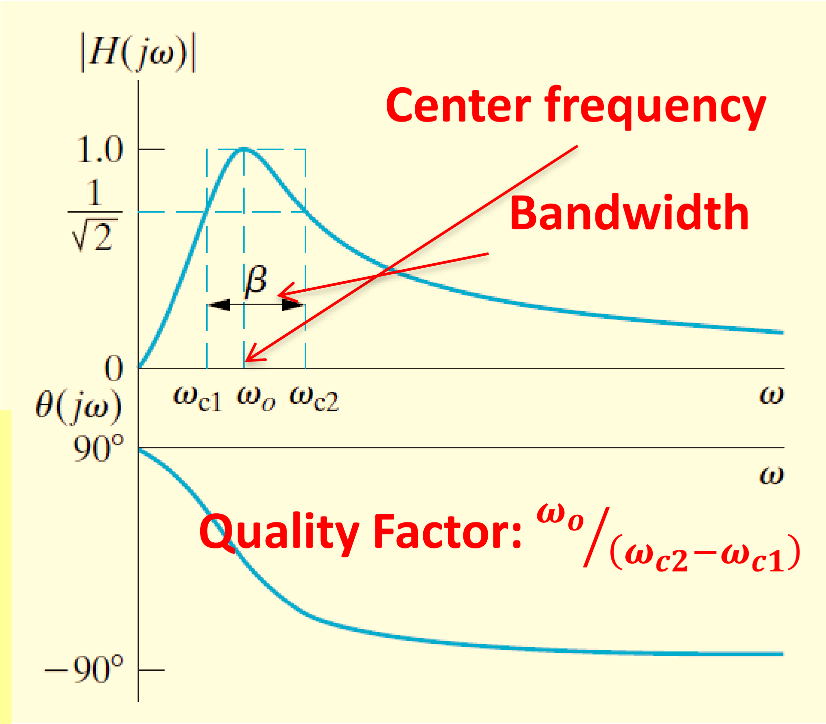
Block

Band-Pass Filters

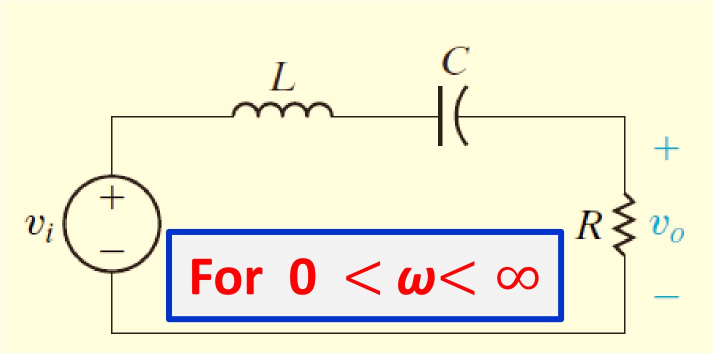
Are those who pass voltages within a band of frequencies to the output while filtering out voltages at frequencies outside this band

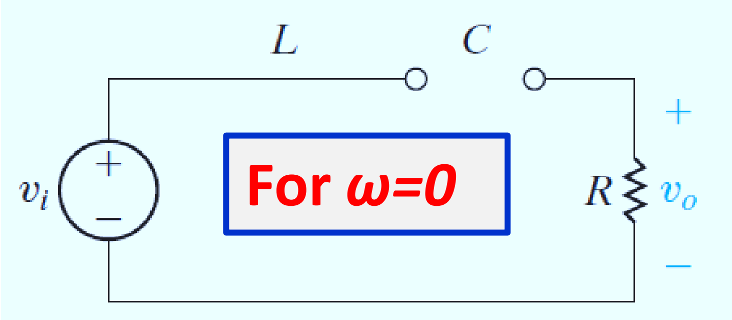


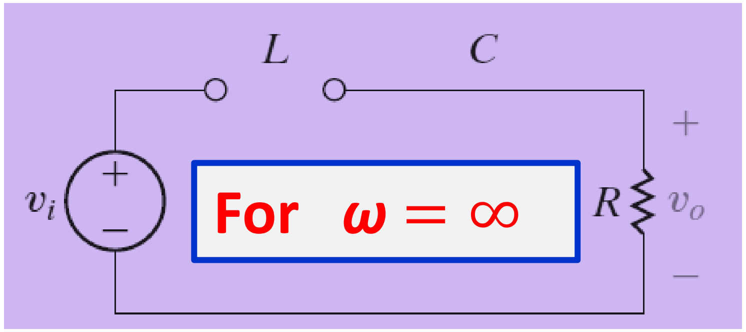
Cutoff frequencies are defined as the ones for which the magnitude of the transfer function equals $^1/_{\sqrt{2}} H_{max}$

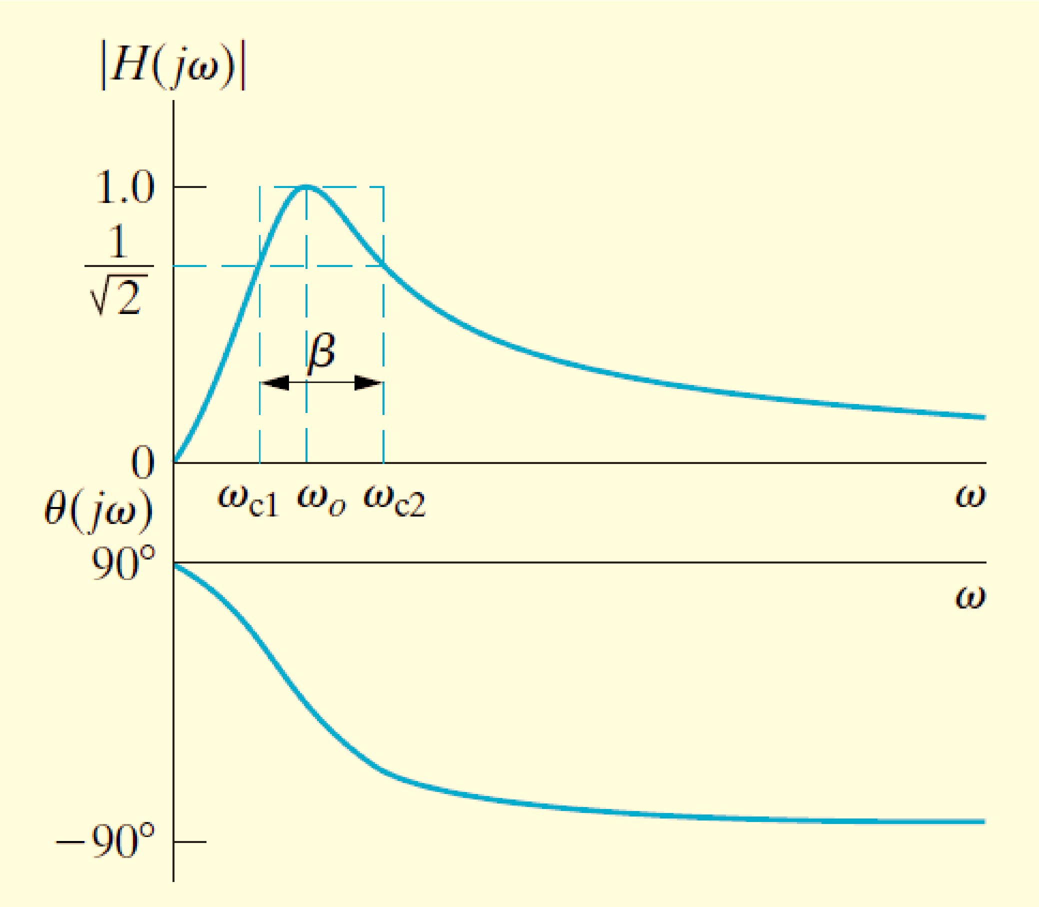


Series RLC Circuit

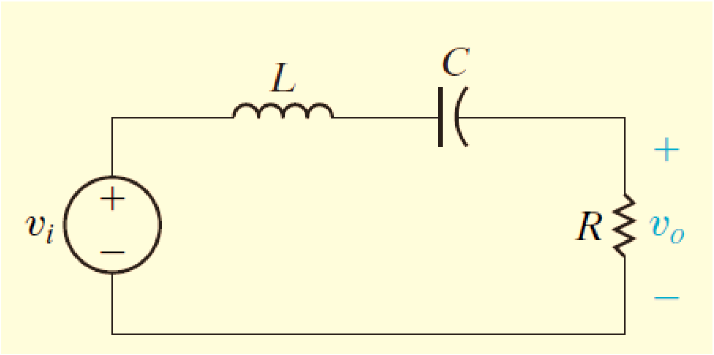








$0 < \omega < \infty$



Between these two extremes, the capacitor C and the inductor L have finite impedances.

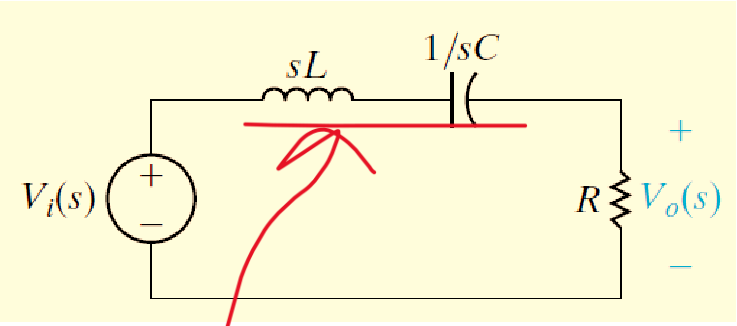
- Voltage supplied by the source drops across L and C and some voltage will reach the resistor.
- *Zc* is negative, *ZL* is positive: at some frequency, they are equal and opposite and cancel out: output voltage = source voltage.

$$Z_{series}(j\omega) = \underbrace{j\omega L}_{Z_L} + \frac{1}{\underbrace{j\omega C}_{Z_C}}$$

This special frequency is the center frequency

The series combination of L and C appears as a short circuit.

Transfer Function



$$j\omega_o L + \frac{1}{j\omega_o C} = 0.$$

$$\omega_o = \sqrt{\frac{1}{LC}}.$$

$$H(s) = \frac{(R/L)s}{s^2 + (R/L)s + (1/LC)}.$$

$$|H(j\omega)| = \frac{\omega(R/L)}{\sqrt{[(1/LC) - \omega^2]^2 + [\omega(R/L)]^2}},$$

$$\theta(j\omega) = 90^\circ - \tan^{-1}\left[\frac{\omega(R/L)}{(1/LC) - \omega^2}\right].$$

The center frequency is defined as the frequency where the transfer function is purely real

Cutoff Frequencies ω_{c1} and ω_{c2}

$$H_{\max} = |H(j\omega_o)|$$

At center frequency H is maximum

$$= \frac{\omega_o(R/L)}{\sqrt{[(1/LC) - \omega_o^2]^2 + (\omega_o R/L)^2}}$$

$$= \frac{\sqrt{(1/LC)}(R/L)}{\sqrt{[(1/LC) - (1/LC)]^2 + \left[\sqrt{(1/LC)}(R/L)\right]^2}} = 1.$$

Cutoff frequencies are defined as the ones for which the magnitufe of the transfer function equals $\mathbf{{}^{1}/_{\sqrt{2}}} \boldsymbol{H_{max}}$

$$\frac{1}{\sqrt{2}} = \frac{\omega_c(R/L)}{\sqrt{[(1/LC) - \omega_c^2]^2 + (\omega_c R/L)^2}} = \frac{1}{\sqrt{[(\omega_c L/R) - (1/\omega_c RC)]^2 + 1}}.$$

Cutoff frequencies ω_{c1}, ω_{c2}

$$\pm 1 = \omega_c \frac{L}{R} - \frac{1}{\omega_c RC}.$$

$$\omega_c^2 L \pm \omega_c R - 1/C = 0.$$

$$\omega_{c1} = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \left(\frac{1}{LC}\right)},$$

$$\omega_{c2} = \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \left(\frac{1}{LC}\right)}.$$

The two positive values of the second order equation have physical meaning and identify the passband of the filter

Center Frequency ω_o

$$\omega_o = \sqrt{\omega_{c1} \cdot \omega_{c2}}$$

The center frequency is the geometric mean of the two cutoff frequencies

$$= \sqrt{\left[-\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \left(\frac{1}{LC}\right)}\right]\left[\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \left(\frac{1}{LC}\right)}\right]}$$

$$= \sqrt{\frac{1}{LC}}.$$

Bandwidth

$$\beta = \omega_{c2} - \omega_{c1}$$

$$= \left[\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \left(\frac{1}{LC}\right)}\right] - \left[-\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \left(\frac{1}{LC}\right)}\right]$$

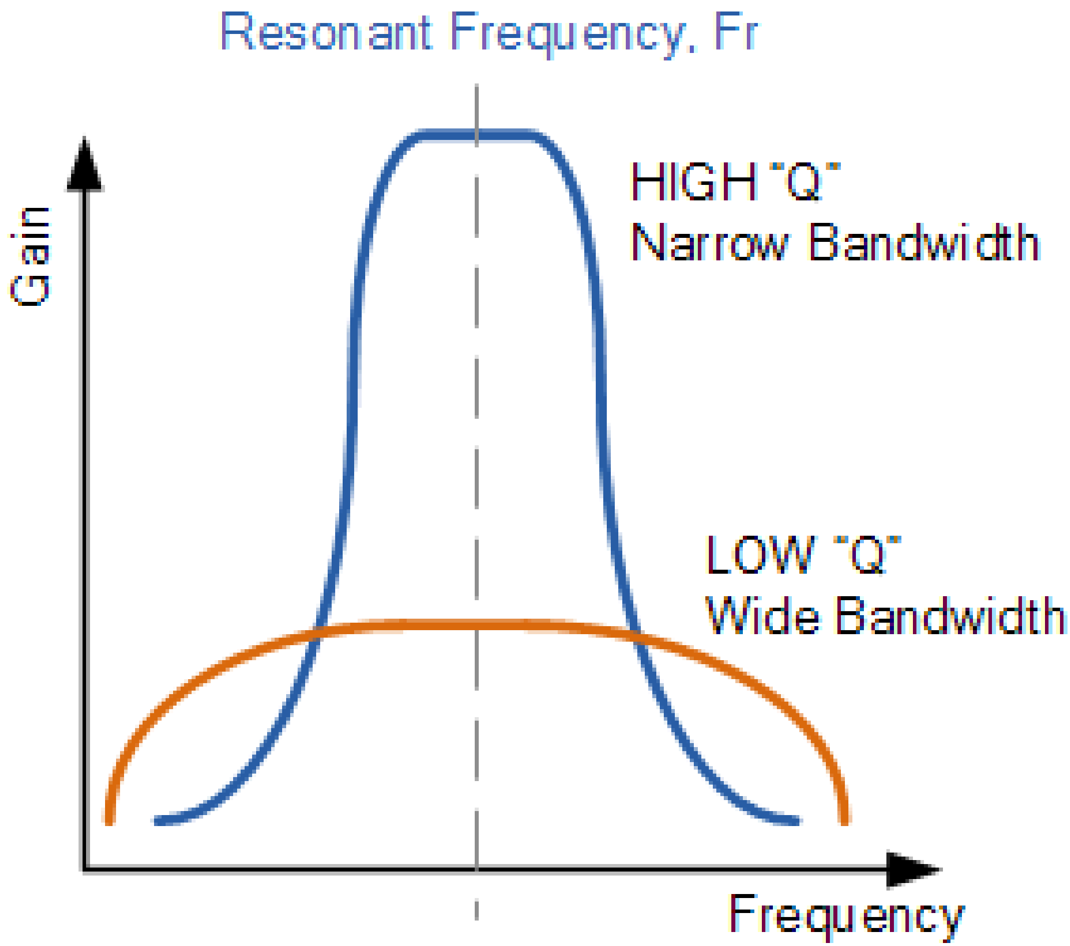
$$= \frac{R}{L}.$$

Cutoff frequencies in terms of the center frequencies and bandwidth

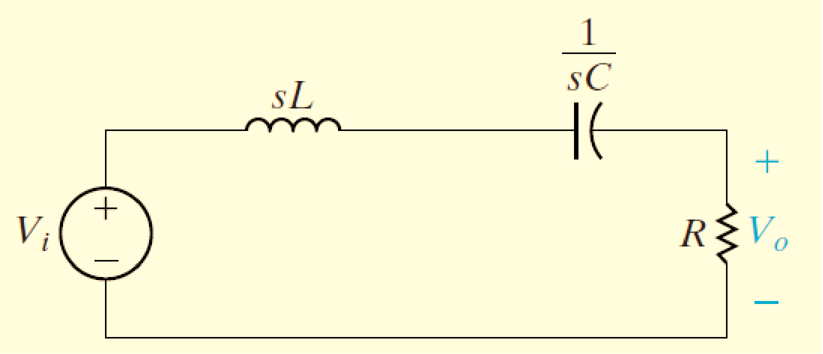
$$\omega_{c1} = -\frac{\beta}{2} + \sqrt{\left(\frac{\beta}{2}\right)^2 + \omega_o^2},$$

$$\omega_{c2} = \frac{\beta}{2} + \sqrt{\left(\frac{\beta}{2}\right)^2 + \omega_o^2}.$$

Q Factor

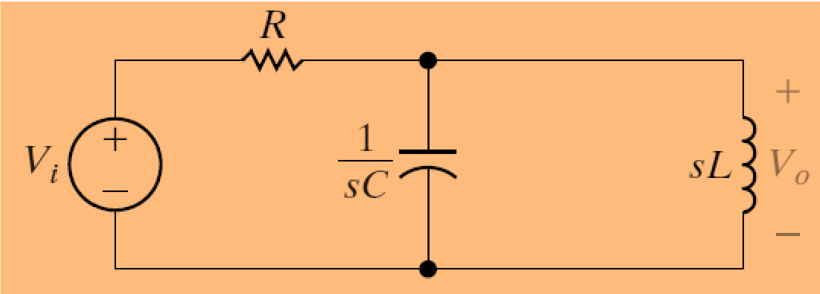

Resonant Frequency, Fr
Gain
HIGH "Q"
Narrow Bandwidth
LOW "Q"
Wide Bandwidth
Frequency

RLC Bandpass Filter



$$H(s) = \frac{(R/L)s}{s^2 + (R/L)s + 1/LC}$$

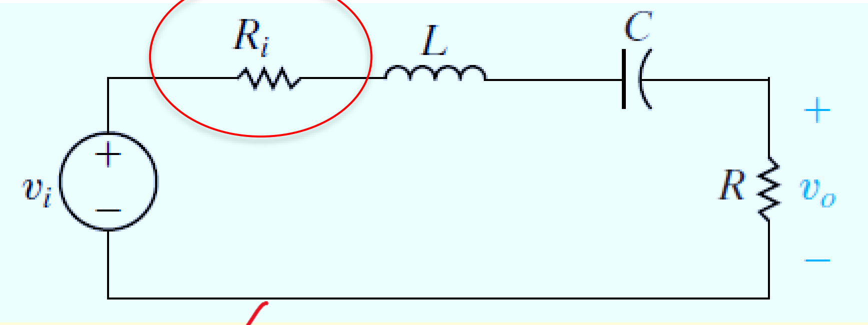
$$\omega_o = \sqrt{1/LC} \qquad \beta = R/L$$



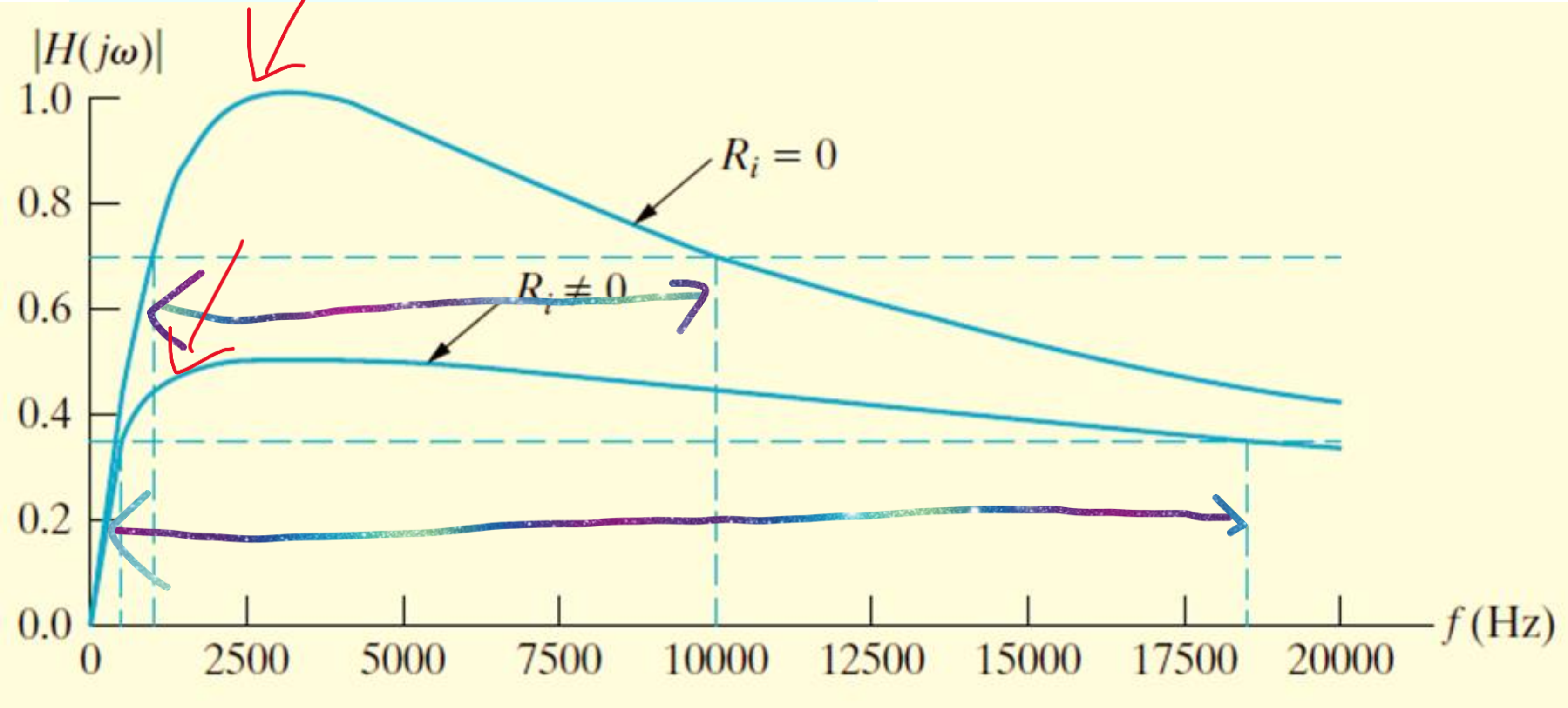
$$H(s) = \frac{s/RC}{s^2 + s/RC + 1/LC}$$

$$\omega_o = \sqrt{1/LC} \qquad \beta = 1/RC$$

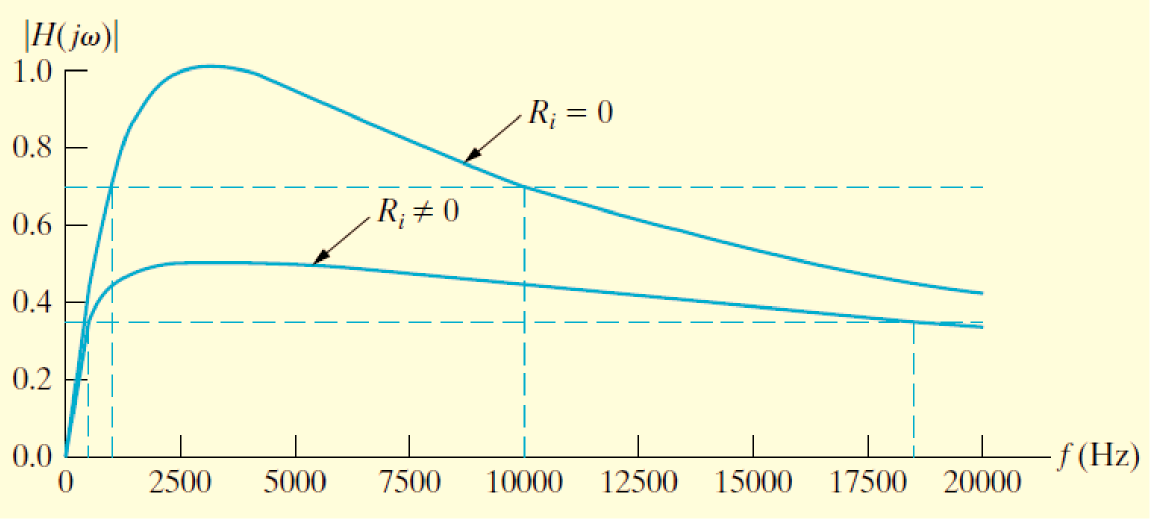
Effect of non ideal voltage source



$$H(s) = \frac{\beta s}{s^2 + \beta s + \omega_o^2}.$$



Effect of non ideal voltage source



- The maximum transfer function magnitude for the filter with $\boldsymbol{R_i \neq 0}$ is smaller than for the filter with $\boldsymbol{R_i = 0}$
- The bandwidth for the filter with $\boldsymbol{R_i \neq 0}$ is larger than that for the filter with $\boldsymbol{R_i = 0}$
- The cutoff frequencies for the two circuits are also different.

Transfer Function

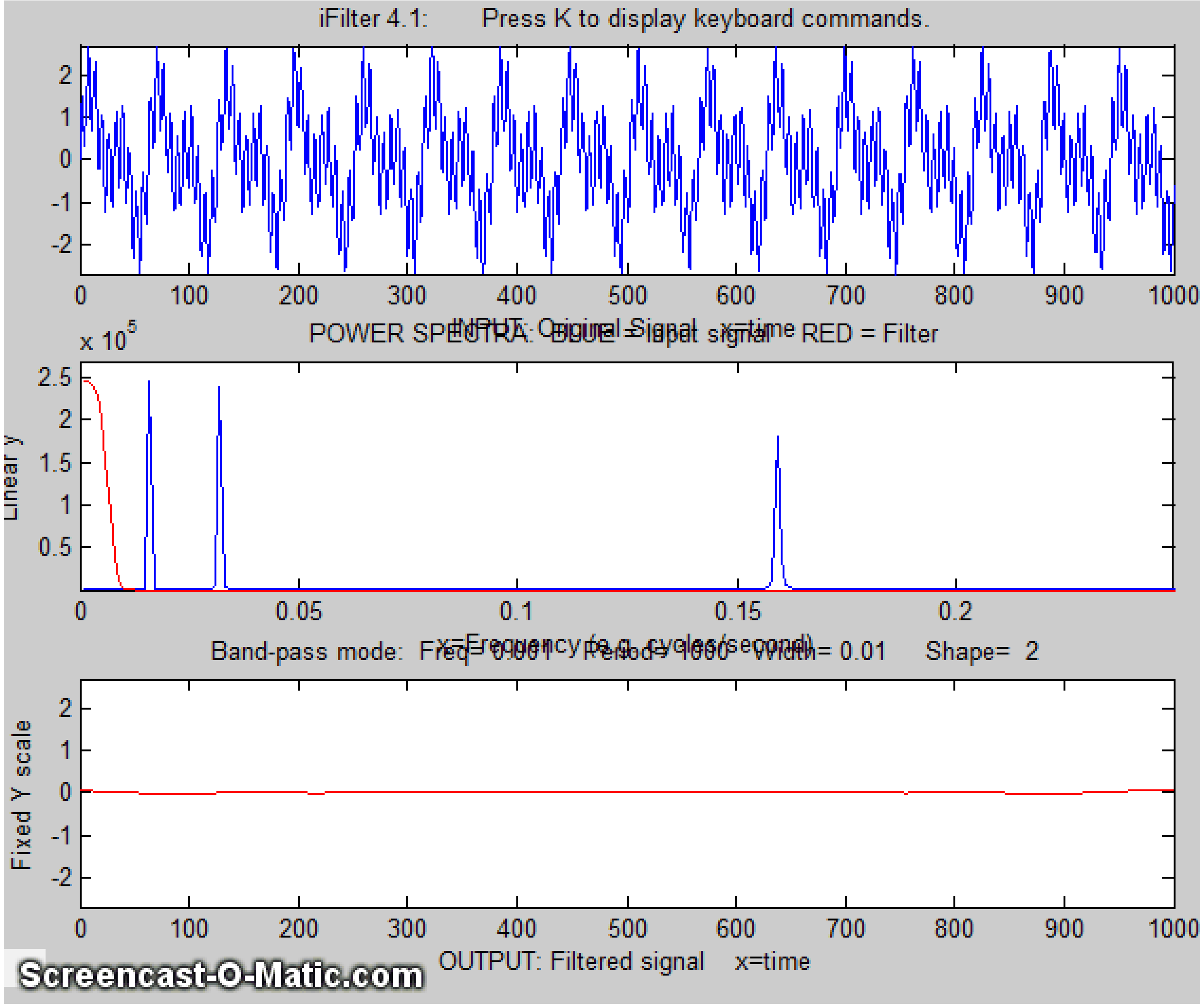
$$H(s) = \frac{\beta s}{s^2 + \beta s + \omega_o^2}.$$

Effect of non ideal voltage source

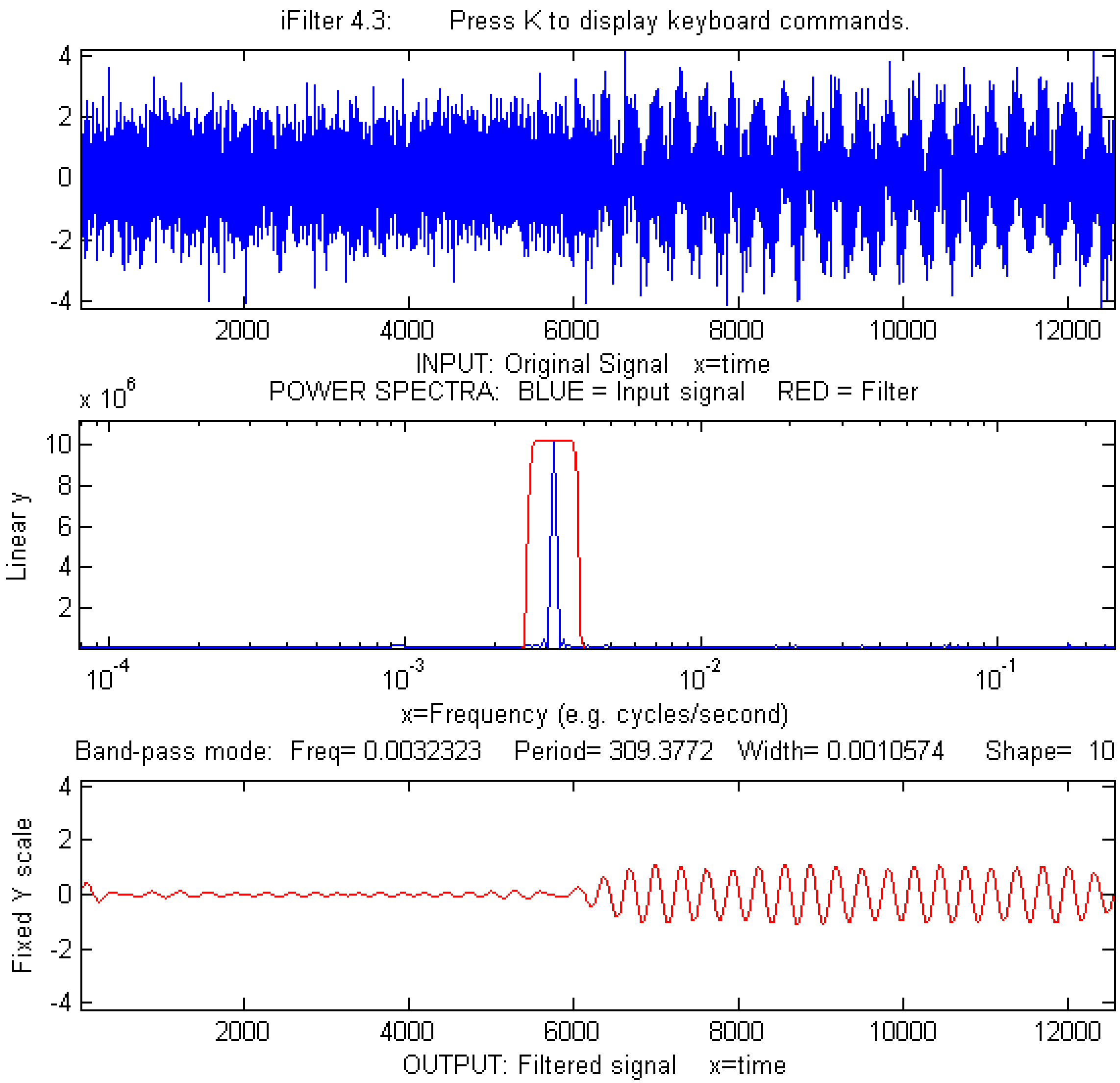
$$H(s) = \frac{K\beta s}{s^2 + \beta s + \omega_o^2},$$

Where K and β depend on whether the series resistance of the voltage source is zero or non zero

Examples



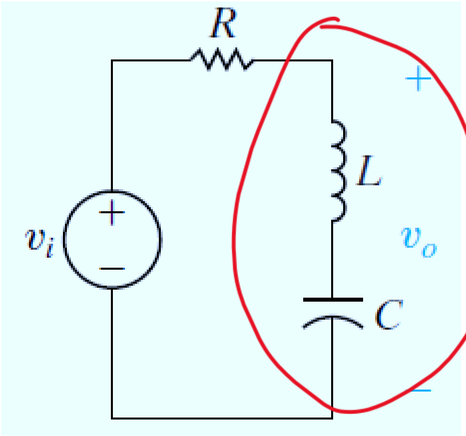
iFilter 4.1: Press K to display keyboard commands.
INPUT: Original Signal x=time
POWER SPECTRA: BLUE = Input signal RED = Filter
Linear y
x=Frequency (e.g. cycles/second)
Band-pass mode: Freq= 0.001 Period= 1000 Width= 0.01 Shape= 2
Fixed Y scale
OUTPUT: Filtered signal x=time
Screencast-O-Matic.com

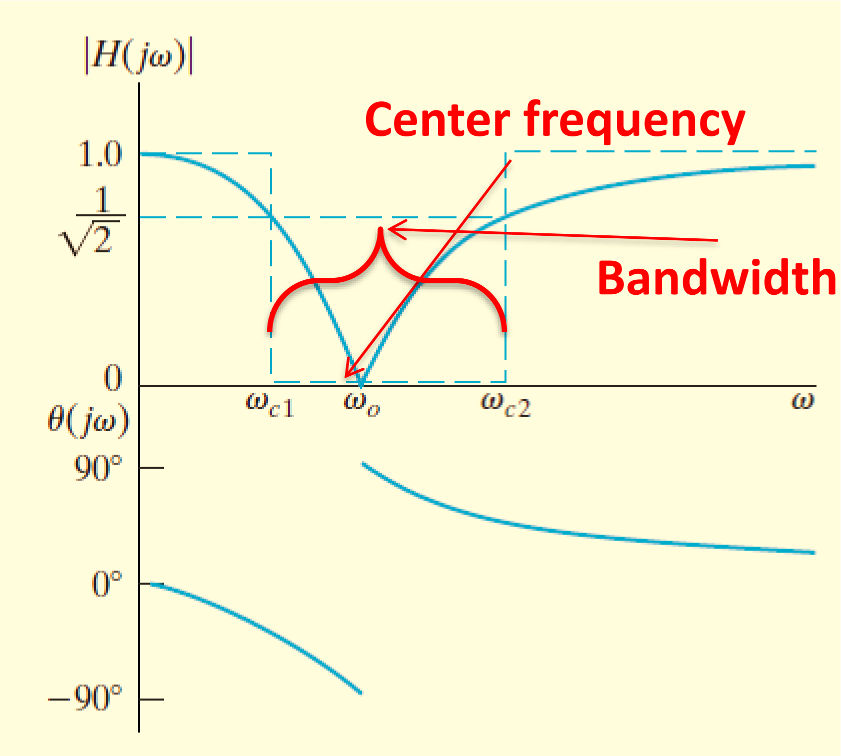

iFilter 4.3: Press K to display keyboard commands.
4
2
0
-2
-4
2000
4000
6000
8000
10000
12000
INPUT: Original Signal x=time
POWER SPECTRA: BLUE = Input signal RED = Filter
x 10^6
Linear y
10
8
6
4
2
10^-4
10^-3
10^-2
10^-1
x=Frequency (e.g. cycles/second)
Band-pass mode: Freq= 0.0032323 Period= 309.3772 Width= 0.0010574 Shape= 10
Fixed Y scale
4
2
0
-2
-4
2000
4000
6000
8000
10000
12000
OUTPUT: Filtered signal x=time

Band-reject filter

Band-Reject Filters

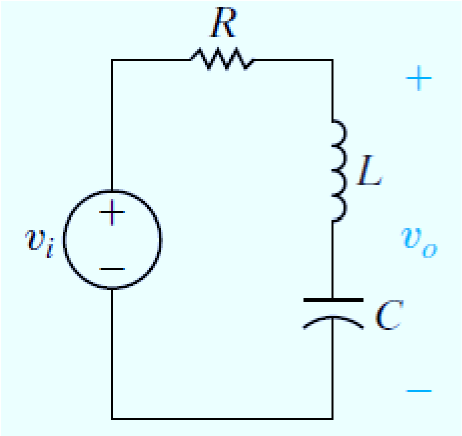
Are those which pass source voltages outside the band between the two cutoff frequencies to the output and attenuates source voltages between the two cutoff frequencies

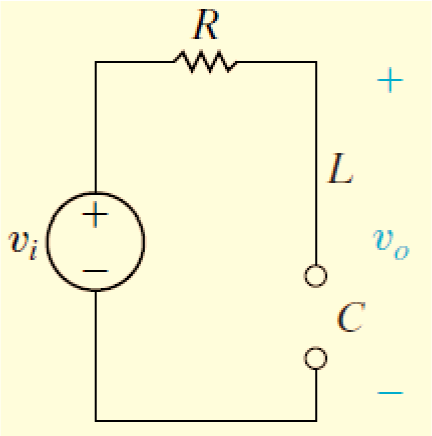




The output voltage is now defined accross the inductor-capacitor pair

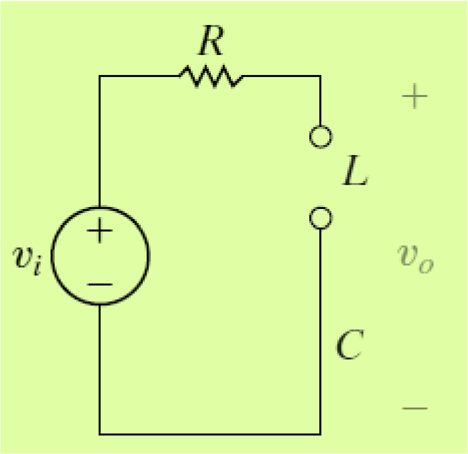
Bandreject Filters





For $0 < \omega < \infty$

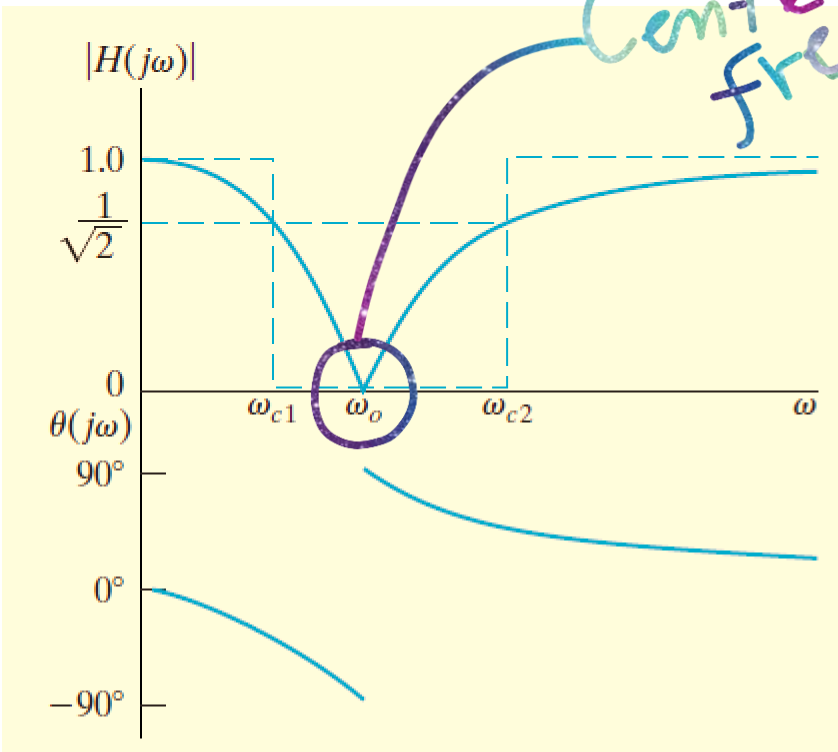
For $\omega = 0$

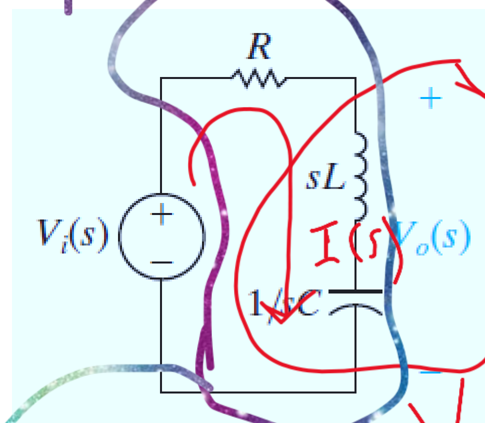


For $\omega = \infty$

Series RLC Circuit

Center frequency





Vout

$$H(s) = \frac{sL + \frac{1}{sC}}{R + sL + \frac{1}{sC}} = \frac{s^2 + \frac{1}{LC}}{s^2 + \frac{R}{L}s + \frac{1}{LC}}.$$

Vin

Between the two passbands, inductor and capacitor have finite impedances of opposite signs.

$$Z_{series}(j\omega) = \underbrace{j\omega L}_{Z_L} + \underbrace{\frac{1}{j\omega C}}_{Z_C}$$

As frequency increases from zero, the impedance of the inductor increases and that of the capacitor decreases

Handnote

Center Frequency ω_o

$$H_{\max} = |H(j0)| = |H(j\infty)|,$$

$$\omega_o = \sqrt{\frac{1}{LC}}.$$

The center frequency is the frequency for which the sum of the impedances of the capacitor and inductor is zero

$$\omega_{c1} = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}},$$

$$\omega_{c2} = \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}.$$

$$Zseries(j\omega) = \underbrace{j\omega L}_{Z_L} + \frac{1}{\underbrace{j\omega C}_{Z_C}}$$

At some frequency between the two passbands, the impedances of the inductor and capacitor are equal but of opposite sign. At this frequency, the series combination of L and C is a short circuit: output voltage is zero. This is the center frequency of this series *RLC* bandreject filter.

Bode plot values: Magnitude & Phase

Handnote

$$|H(j\omega)| = \frac{\left|\frac{1}{LC} - \omega^2\right|}{\sqrt{\left(\frac{1}{LC} - \omega^2\right)^2 + \left(\frac{\omega R}{L}\right)^2}},$$

$$\omega_o = \sqrt{\frac{1}{LC}}.$$

$$\theta(j\omega) = -\tan^{-1}\left(\frac{\frac{\omega R}{L}}{\frac{1}{LC} - \omega^2}\right).$$

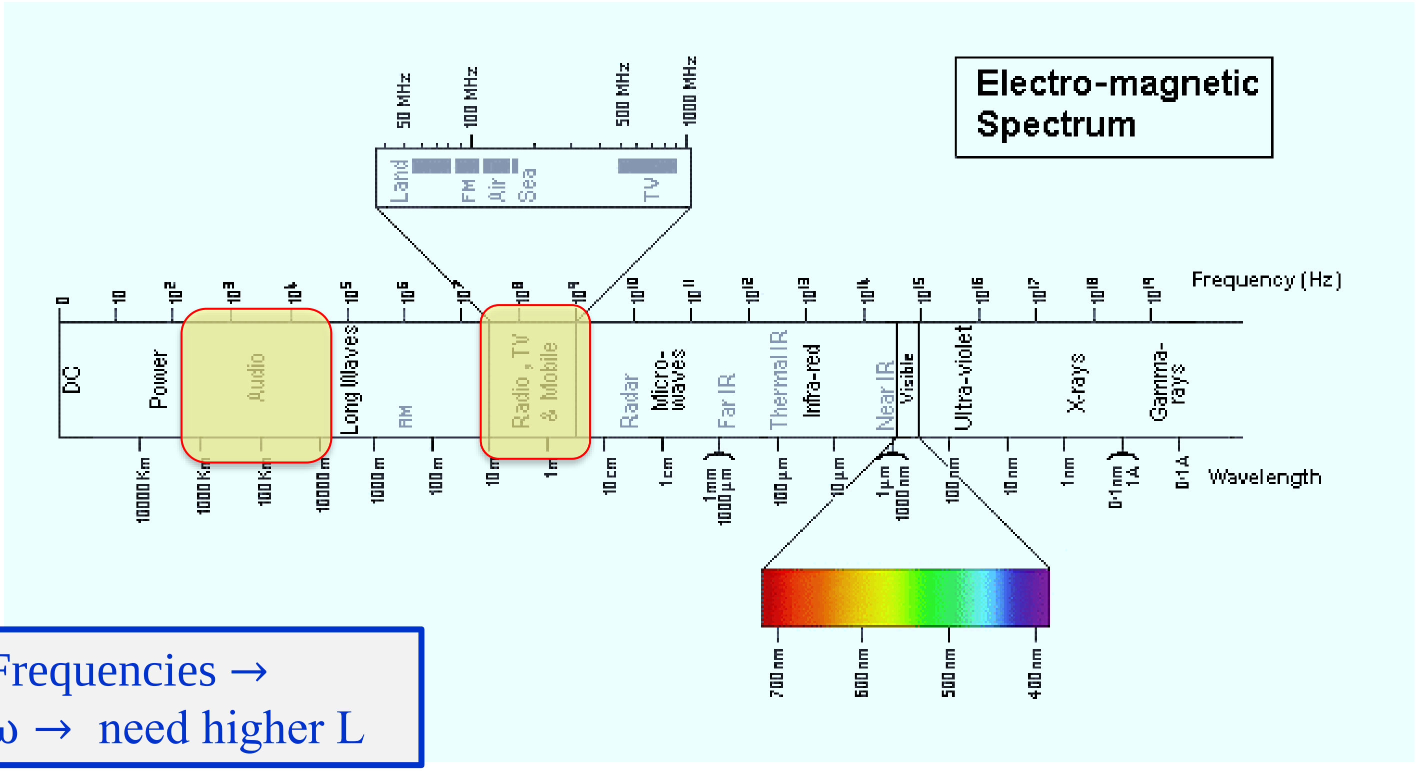
The phase shift between the input and the output approaches **-90°** as $\boldsymbol{\omega L}$ approaches $^1/_{\omega C}$. As soon as $\boldsymbol{\omega L}$ **e**xceeds $^1/_{\omega C}$ the phase shift jumps to **+90°** and then approaches zero as $\boldsymbol{\omega}$ continues to increase.

Loaded
& non-ideal sources filters

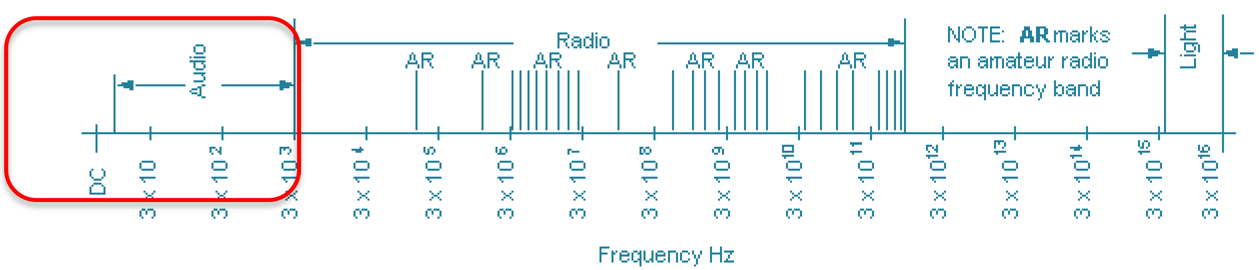
Load: Adding a load to the output of a passive filter changes its filtering properties by altering the location and magnitude of the passband.

Non-ideal sources: Replacing an ideal voltage source with one whose source resistance is nonzero also changes the filtering properties of the rest of the circuit, again by altering the location and magnitude of the passband.

Audio & Radio Frequencies



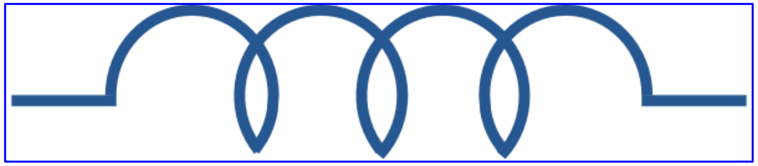
Audio Frequencies →
lower ω → need higher L



Passive Filters: Review

- Made up of passive components - resistors, capacitors and inductors
- No amplifying elements (transistors, op-amps, etc)
- No signal gain
- Require no power supplies
- Desirable to use inductors with high quality factors $\left[Q = \frac{\omega L}{R}\right]$

$$X_L = \omega L$$



Audio Frequencies: bulky inductors
Radio Frequencies: OK

End